

Arnab Bhowmik

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EDUCATION

Stony Brook University

May 2027

Bachelor of Science with Honors in Computer Science

GPA: 3.81

Relevant Coursework: Software Development, Software Engineering, Theory of Computation: Honors, Analysis of Algorithms: Honors, Data Structures, Object-Oriented Programming, Systems Programming, Programming Abstractions, Machine Learning

EXPERIENCE

CoreWeave

Jun. 2026 – Aug. 2026

Software Engineering Intern, Cloud Platform API/IAM

New York, NY

GlaxoSmithKline

May 2025 – Aug. 2025

Compute Platform Engineering Intern

Seattle, WA

- Developed an interactive **Python CLI** that uses workload diagnosis to auto-select optimal HPC environments across **Slurm** and **Google Batch** and optimize resource specifications when applicable, with two submission modes (automatic job script generation/submission and direct environment access), **reducing compute costs by ~10%**.
- Containerized and deployed the CLI using both **Docker** and **Apptainer** for cross-platform compatibility on Windows, Linux, and Unix systems, with planned rollout to **3,000+** computational scientists company-wide.
- Built proof of concept demonstrating architectural optimizations for AI/ML team's prototype agentic system's tool orchestration layer, achieving **~35% reduction in context usage**.

Stony Brook University

Jan. 2025 – Dec. 2025

Teaching Assistant

Stony Brook, NY

- Led recitations, review sessions, and office hours for **250+** students across Programming Abstractions (CSE 216) and Software Development (CSE 316).
- Helped revise course materials, graded assignments/exams, and proctored exams to ensure smooth course operations.

Stony Brook University

Sep. 2024 – Present

Student Software Developer, Vertically Integrated Projects (VIP) Program

Stony Brook, NY

- Led the HealthByte subteam and its development by delegating tasks, creating onboarding resources, and organizing meetings.
- Developed prototype patient-facing **iOS** and clinician-facing **full-stack web** applications for post-surgery recovery monitoring using **Swift**, **HealthKit**, and **ResearchKit**, with a centralized **PostgreSQL** database across both platforms.
- Developing **Regio Vinco**, an interactive web-based educational game helping primary school students learn world geography, iterating on versions tested with **150+** students at partnering schools.

PenguinLearn

Jul. 2024 – May 2025

Full Stack Developer

Remote

- Led full-stack development of **PenguinLearn**, a **RESTful educational platform**, using Next.js, React, Supabase, and Prisma ORM, enabling migration from third-party hosting and **reducing overall operational costs by ~20%**.
- Implemented an on-site **real-time messaging system** for direct communication between parents and teachers, integrated **Stripe payment processing**, and automated class meeting scheduling via the **Zoom API**.
- Built test suites with **Jest** and **Playwright** and set up a **CI/CD pipeline**, ensuring reliability and streamlined deployments.

PROJECTS

TA Tools | *Python, Flask, Beautiful Soup, Selenium WebDriver, SQLite, JavaScript*

Jul. 2024 – Aug. 2024

- Developed a full-stack web application using Flask, Jinja, and SQLite to **automate logistics tasks** for teaching assistants at a tutoring center, improving task efficiency by approximately **200%** for active users.

Prediction Market Maker Bot | *Python, Redis, PM2, AWS EC2*

Dec. 2025 – Present

- Designed and deployed a **market-making system** for prediction markets, actively trading on **Polymarket US**.
- Implemented automated quoting, inventory management, and risk controls to provide continuous liquidity across markets, processing **1.1M+ trades** and **83M+ contracts** with over **\$34M** in notional volume.
- Achieved profitability at scale on Polymarket, generating **\$375,000+** in profit (**8,500%+ return** on initial investment).
- Built a resilient trading system using **Redis for state management**, **PM2 for process supervision**, and **containerized AWS EC2 deployments**, with **email alerts** for stats and system health.

SKILLS

Languages: Java, Python, Go, SQL, Bash, C, OCaml, JavaScript, PHP, Swift

Frameworks: NumPy, pandas, React, Vue, LangChain, PyTorch, scikit-learn, Next.js, Express.js, Node.js, FastAPI

Tools: PostgreSQL, MySQL, SQLite, MongoDB, Redis, Pinecone, Git, Docker, Kubernetes, GitHub Actions, Ansible, Terraform, PM2, Jira, Slurm, Google Cloud Platform (GCP), Amazon Web Services (AWS)

Concepts: Distributed Systems, RESTful APIs, Agile Development, Machine Learning, Natural Language Processing (NLP), High Performance Computing (HPC), Linux, Unix, LLMs, Model Context Protocol (MCP)